

Cam Morreale

Reverse-engineers broken data systems into scalable decision infrastructure

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EDUCATION

Worcester Polytechnic Institute — MS Data Science (May 2026), BS Data Science (May 2025)

SKILLS

Programming & Systems — Python, SQL, R, Bash, React, TypeScript, Git

Data Engineering & Architecture — ETL/ELT pipelines, data modeling, data validation, data quality & deduplication, schema design, data ingestion & normalization

Data & Analytics Tooling — Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Jupyter/ipywidgets

Cloud & Data Infrastructure — Snowflake, Snowpark, AWS Lambda, S3, IAM, Airbyte, Docker, Oracle SQL

PROFESSIONAL EXPERIENCE

Wellington Management — Data Architecture & Engineering Intern | Boston, MA June–Aug 2025

- Reconstructed the data architecture powering a \$153B investment platform used by ~160 investment professionals, replacing a fragmented legacy Excel workflow and driving ~\$8M in annual efficiency gains.
- Reverse-engineered an unmapped 250-table Oracle database, inferring schema, primary/foreign key relationships, and data dependencies to reconstruct the system architecture.
- Diagnosed and resolved a missing upstream dependency causing system-wide pipeline failure.
- Co-designed and implemented an analytics-override framework (Python, Pandas, ipywidgets, Oracle SQL), enabling validated, auditable model adjustments for investment workflows.

Cooper.ai — Data Engineer (Contract) | Remote June 2026–Present

- Automated recurring client data-ingestion workflows by replacing manual file preparation and merge steps with scheduled AWS Lambda/S3/Snowflake pipelines.
- Built a production retail-data ingestion automation that merges multiple upstream feeds, handles late-arriving files, writes auditable outputs, and triggers downstream Snowflake processing.
- Automated a logistics-reporting pipeline that converts multi-tab Excel reports into Snowflake-ready datasets, loads five production staging tables, and enables daily analytics/AI refreshes.
- Remediated production shipping-ingestion defects across four carriers, removing 195K+ duplicate FedEx rows, restoring full date coverage, and preventing recurring data corruption.

MassDEP — Technical Business Analyst (Data & AI Systems) | Worcester, MA Jan–May 2025

- Owned retrieval, extraction, and validation layers of an LLM pipeline, converting unstructured reports into Power BI–ready datasets and reducing manual reporting effort by ~40% across 50+ projects.
- Designed and implemented a RAG system to ground outputs in source documents and reduce hallucination.
- Standardized model outputs into structured CSV datasets and integrated them into power BI, enabling downstream analysis of extracted environmental data across documents.
- Built page-level inspection and validation logic to detect unreadable outputs, surface failure states, and quantify system reliability (97% successful extraction rate).

Embue — Data Systems Architecture Lead | Worcester, MA Aug–Dec 2022

- Solved lack of tamper-evident storage in IoT systems by architecting a distributed data infrastructure to enable verifiable data integrity across decentralized environments.
- Designed a layered architecture separating encryption, storage, verification, and access control, improving scalability, security, and system modularity.
- Identified scalability limitations of on-chain storage and implemented a hybrid architecture using IPFS for off-chain storage and Filecoin for verification, balancing performance with integrity guarantees.
- Built and shipped an end-to-end encrypted pipeline (GnuPG → IPFS → Filecoin), enabling secure ingestion, storage, and retrieval of IoT data.